

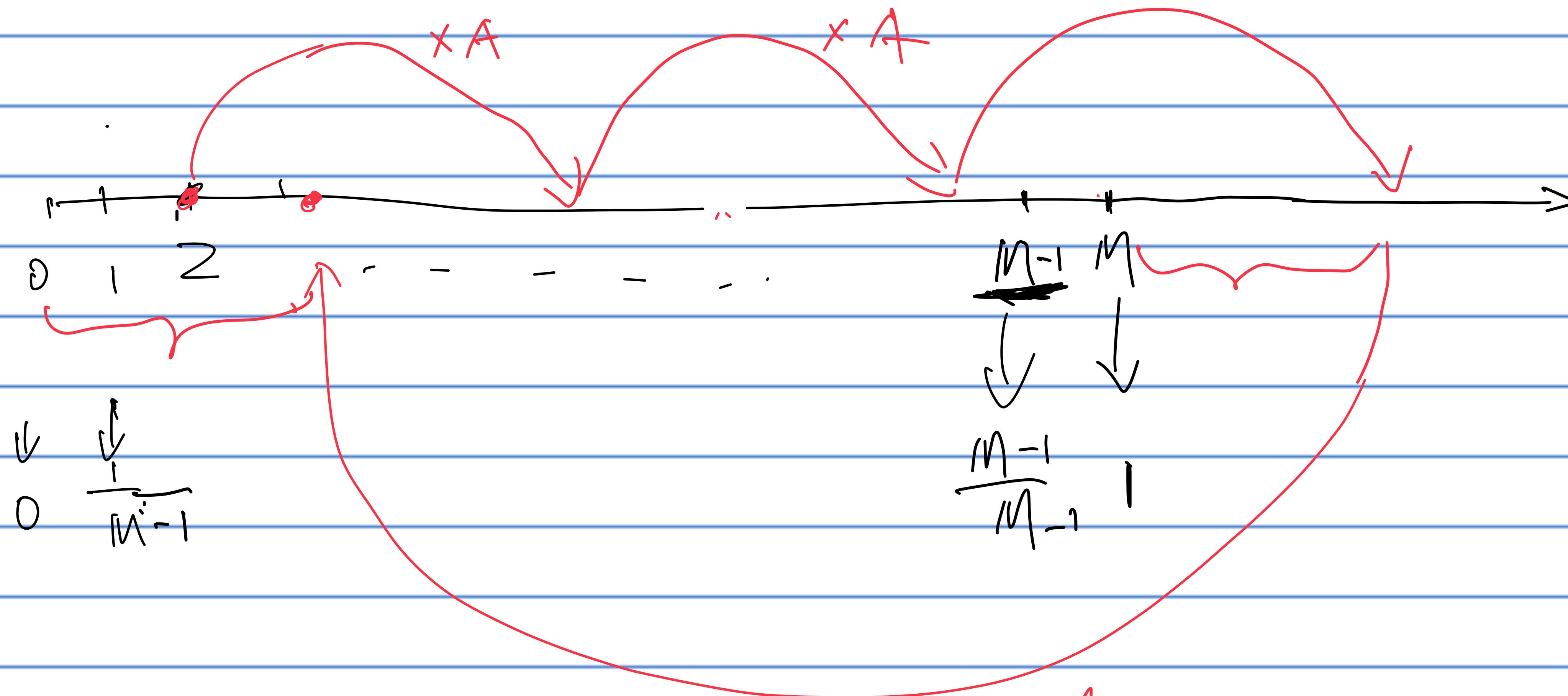
STAT 420/812

Computational Statistics

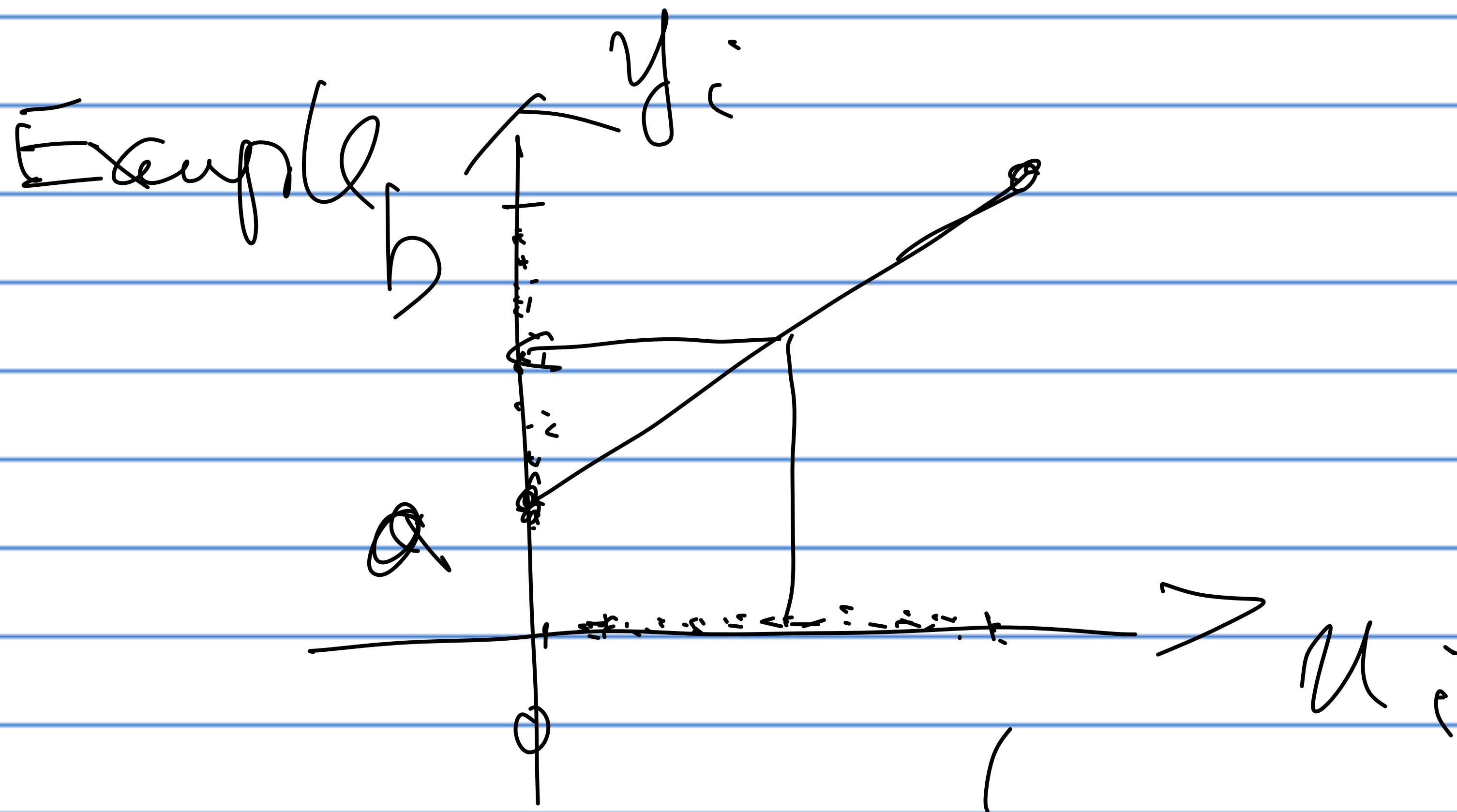
Simple Monte Carlo

Longhai Li, 2022

Pseudo Random Numbers.



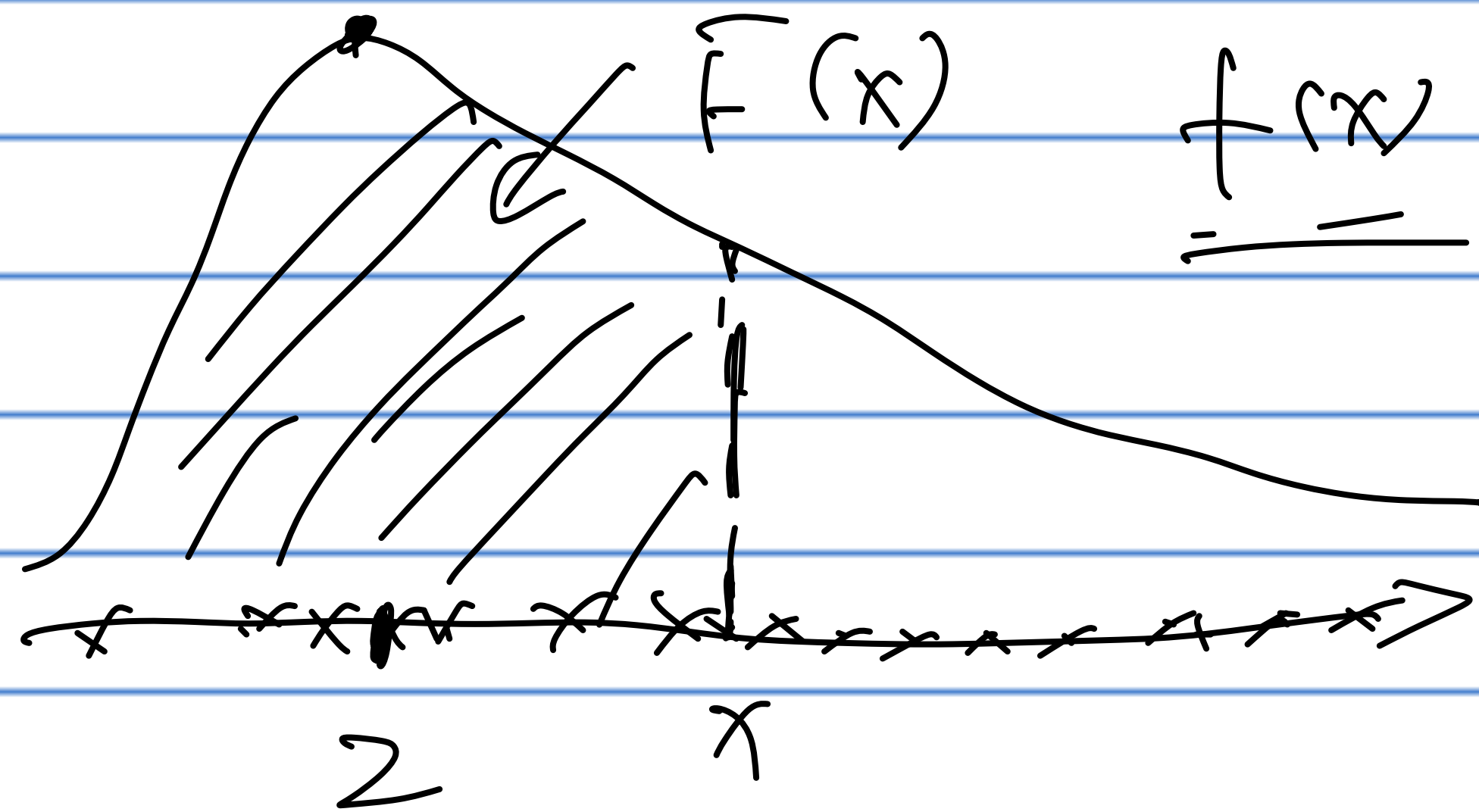
$$a_{i+1} = (A \cdot a_i) \text{ mod } M, \text{ set seed (100)}$$



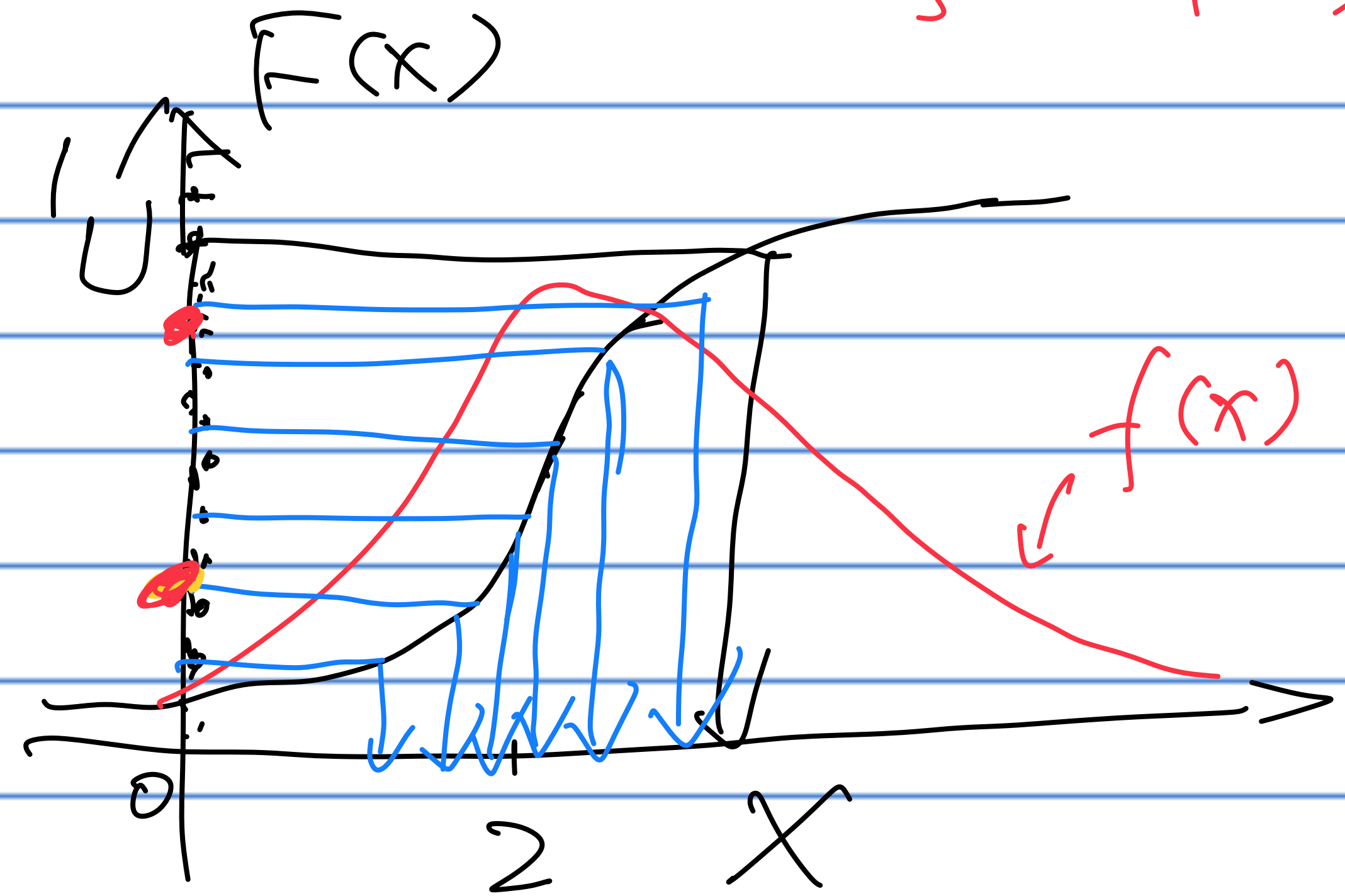
$$y_i = a + (b-a) \cdot u_i$$

Inverting C.D.F. (Continuous case)

X has a p.d.f. f



$$f(x) = F'(x)$$



$$F(x) = \int_{-\infty}^x f(t) dt$$

$$X = F_x^{-1}(U)$$

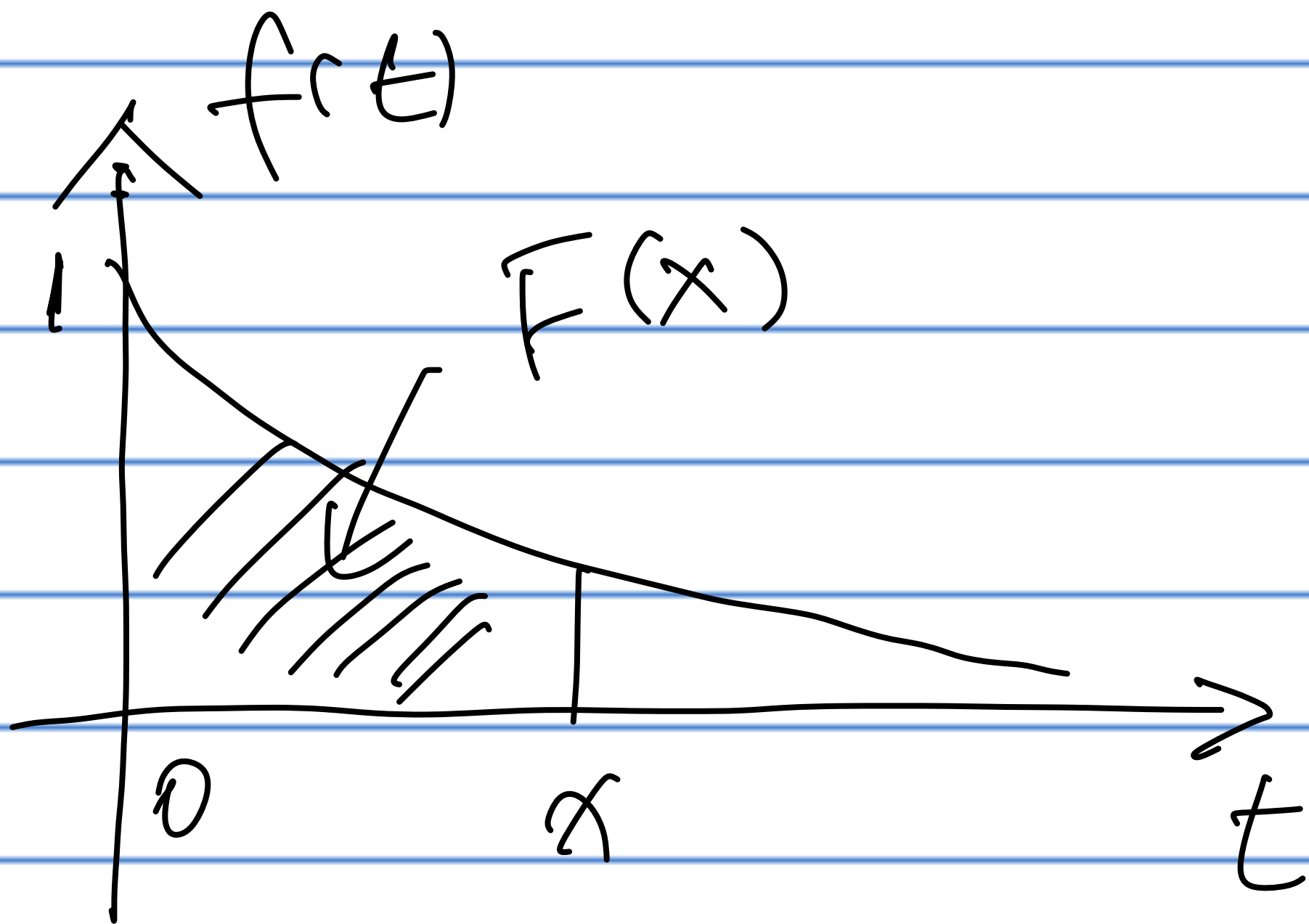
$$\text{Pf } \dot{X} = F_x^{-1}(U), \quad U = F_x(x)$$

$$P(X \leq x) = P(F_x^{-1}(U) \leq x) \\ = P(U \leq F_x(x))$$

$$f_x(x) = \int_U^{F_x(x)} \left| \frac{du}{dx} \right|_+ = f_U(F_x(x)) \cdot f(x) \\ = 1 \cdot f(x)$$

Example:

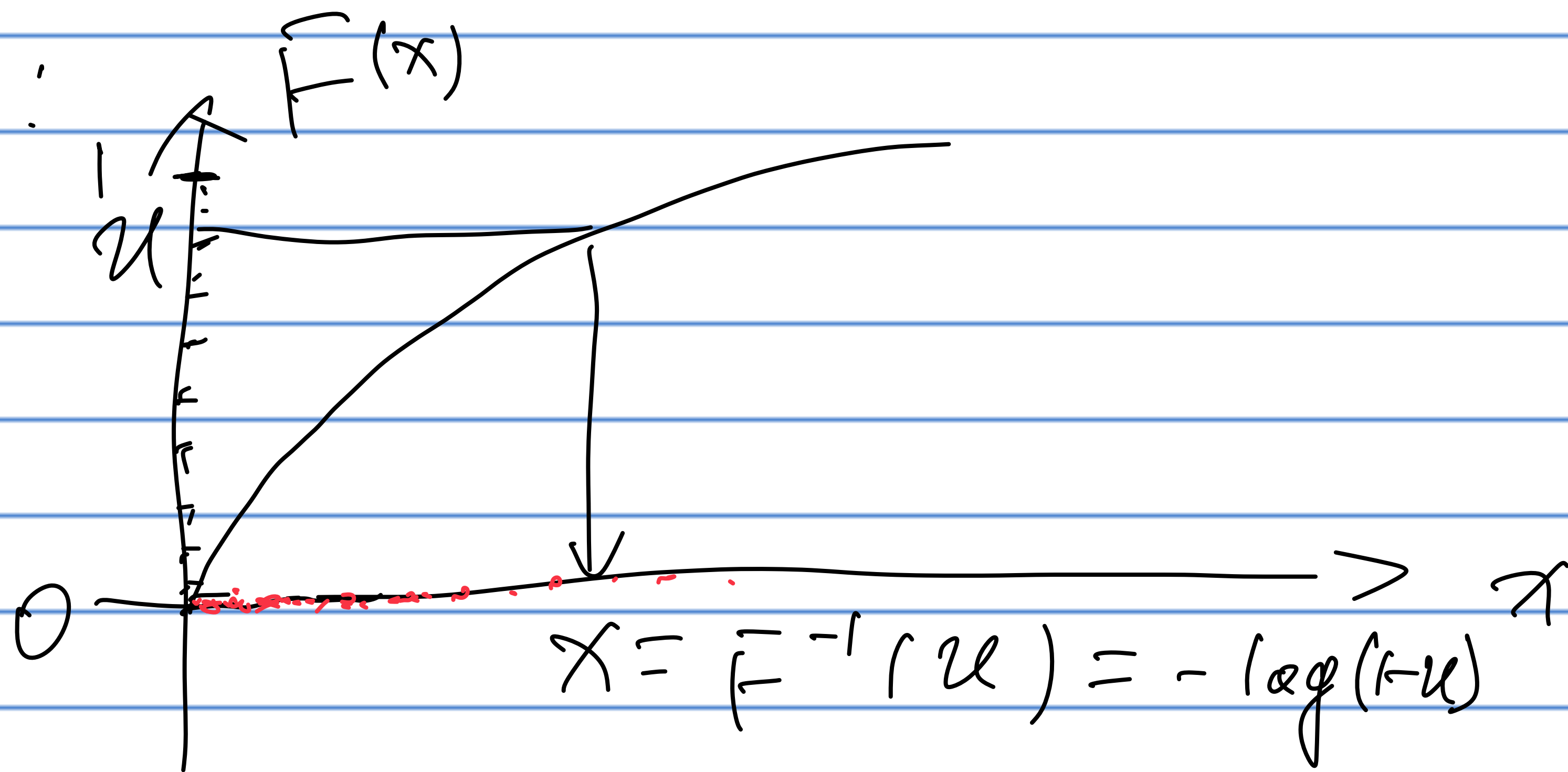
$$f(x) = e^{-x}, \text{ for } x > 0$$



$$F(x) = \int_0^x f(t) dt$$

$$= \int_0^x e^{-t} dt$$

$$= \left. -e^{-t} \right|_{t=0}^x$$
$$= 1 - e^{-x}$$

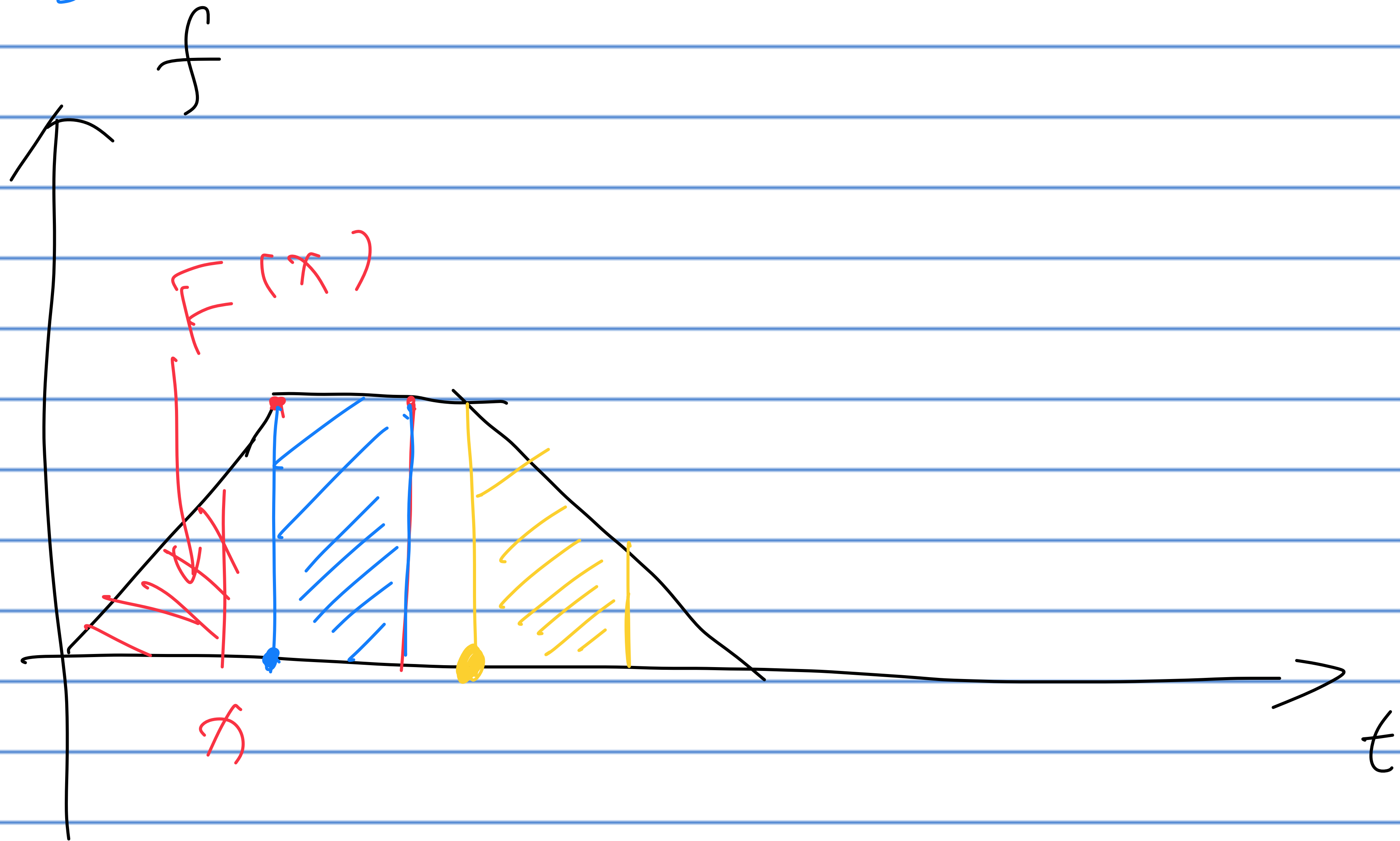


$$F(x) = 1 - e^{-x}, \quad \text{for } x > 0$$

$$u = 1 - e^{-x}$$

$$x = -\log(1-u) \equiv F^{-1}(u)$$

Assignment 1 Q2 :



Generating $N(0,1)$ r.n.

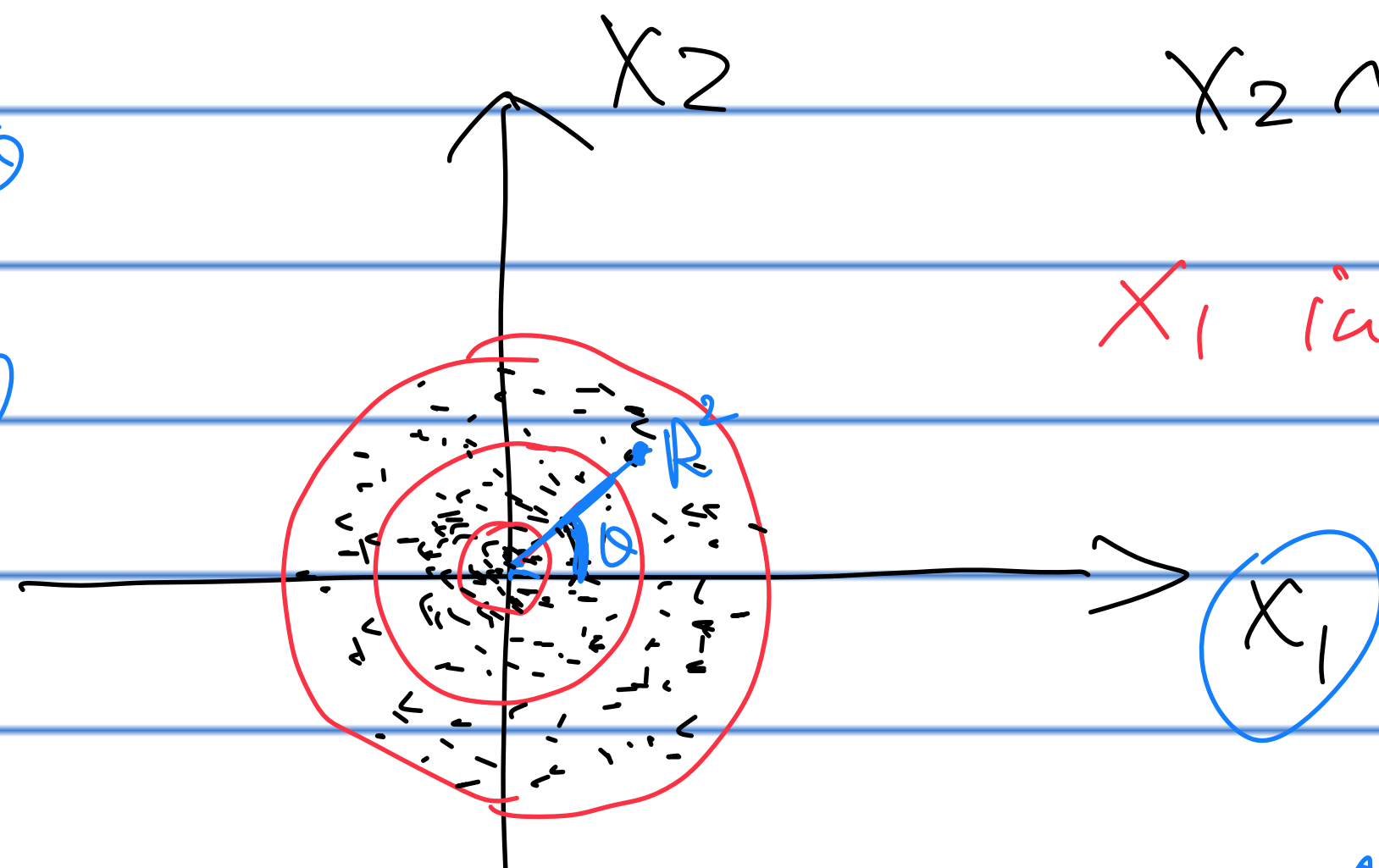
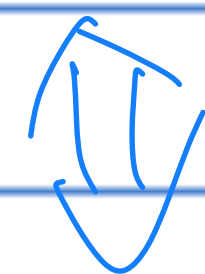
$$X_1 \sim N(0,1)$$

$$X_2 \sim N(0,1)$$

$$f(x_1, x_2)$$

X_1 indep X_2

$$\begin{cases} X_1 = R \cdot \cos \theta \\ X_2 = R \cdot \sin \theta \end{cases}$$



$$= \frac{1}{2\pi} e^{-\frac{x_1^2 + x_2^2}{2}}$$

$$\begin{cases} R^2 = X_1^2 + X_2^2 \sim \chi^2_2 = \text{Gamma}(1, \beta=2) \\ = \text{Exp}(\beta=2) \end{cases}$$

$$\begin{cases} \theta = \text{angle of } (X_1, X_2) \text{ to } X_1 \text{ axis} \sim \text{Unif}(0, 2\pi) \end{cases}$$

$$f(x_1, x_2) = \frac{1}{2\pi} e^{-\frac{x_1^2 + x_2^2}{2}}$$

Theorem 5.1.1 (Strong Law of Large Number) *If random variables $X_1, X_2, \dots$, iid from a distribution with finite mean μ , then*

$$P\left(\frac{X_1 + \dots + X_n}{n} \rightarrow \mu\right) = 1 \tag{5.1}$$

This theorem says that if we can simulate X_i long enough, the sample average will almost surely converge to the true mean.

Theorem 5.1.2 (Central Limit Theorem) *If random variables $X_1, X_2, \dots$, iid from a distribution with finite mean μ and finite standard deviation σ , then*

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \rightarrow^d N(0, 1) \quad (5.2)$$

CLT can be used to calculate error bound given a confidence level. For instance, given confidence level $\alpha = 95\%$, we can conclude from CLT:

$$P(-1.96 \sigma/\sqrt{n} \leq \bar{X} - \mu \leq 1.96 \sigma/\sqrt{n}) \approx 0.95 \quad (5.3)$$

Obviously, we see that the width of error bound will decrease with n , in a rate of square root.

$$\begin{aligned} -1.96 < \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} < 1.96 &\Leftrightarrow |\bar{X} - \mu| < 1.96 \cdot \frac{\sigma}{\sqrt{n}} \\ \Leftrightarrow \bar{X} - 1.96 \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + 1.96 \frac{\sigma}{\sqrt{n}} \end{aligned}$$

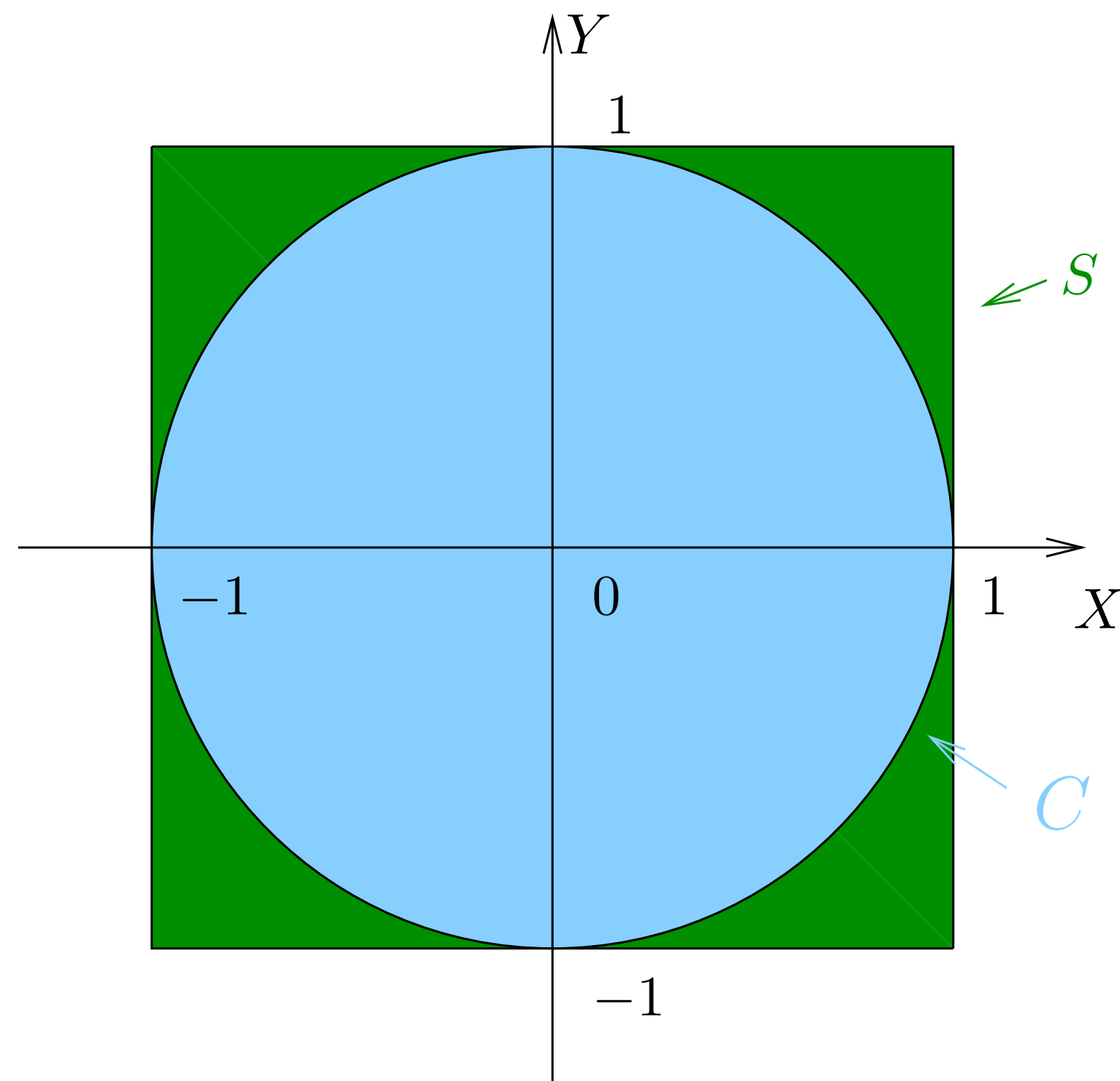
A toy example: estimating π with Monte Carlo method

Suppose random variables X, Y are uniformly distributed over rectangle $S = (-1, 1) \times (-1, 1)$, as displayed in Figure 5.2. The probability that (X, Y) lies in the ball $C = \{(x, y) | x^2 + y^2 \leq 1\}$ is $\pi/4$, in other words,

$$E(4 \times I((X, Y) \in C)) = \pi.$$

We can draw samples of (X, Y) uniformly from S and count how much proportion of these samples fall into C , then π is estimated by:

$$\hat{\pi}_n = 4 \times \frac{\sum_{i=1}^n I(X_i, Y_i) \in C}{n} \tag{5.4}$$



$$P((X, Y) \in C) = \frac{\pi}{4}$$

$$E(I((X, Y) \in C)) = \frac{\pi}{4}$$

$$(X, Y) \sim \text{Unif}(S)$$

The probability that (X, Y) lies in the ball $C = \{(x, y) | x^2 + y^2 \leq 1\}$ is $\pi/4$

$$E(4 \times I((X, Y) \in C)) = \pi$$

$$I((X, Y) \in C) = \begin{cases} 1, & (X, Y) \in C \\ 0, & \text{o.w.} \end{cases}$$

$$Z = 4 \cdot I((X, Y) \in C), E(Z) = \pi$$

$$Z_1, Z_2, \dots, Z_n \Rightarrow \bar{Z} = \frac{Z_1 + \dots + Z_n}{n} = \hat{\pi}$$

Evaluation of Point Estimators

$$\hat{\theta}^{(1)} = \hat{\theta}(x_1^{(1)}, \dots, x_n^{(1)})$$

$$\hat{\theta}^{(2)} = \hat{\theta}(x_1^{(2)}, \dots, x_n^{(2)})$$

$$MSE = E[(\hat{\theta} - \theta)^2]$$

$$MSE = \frac{\sum_{i=1}^N (\hat{\theta}^{(i)} - \theta)^2}{N}$$

$$\hat{\theta}^{(N)} = \hat{\theta}(x_1^{(N)}, \dots, x_n^{(N)})$$

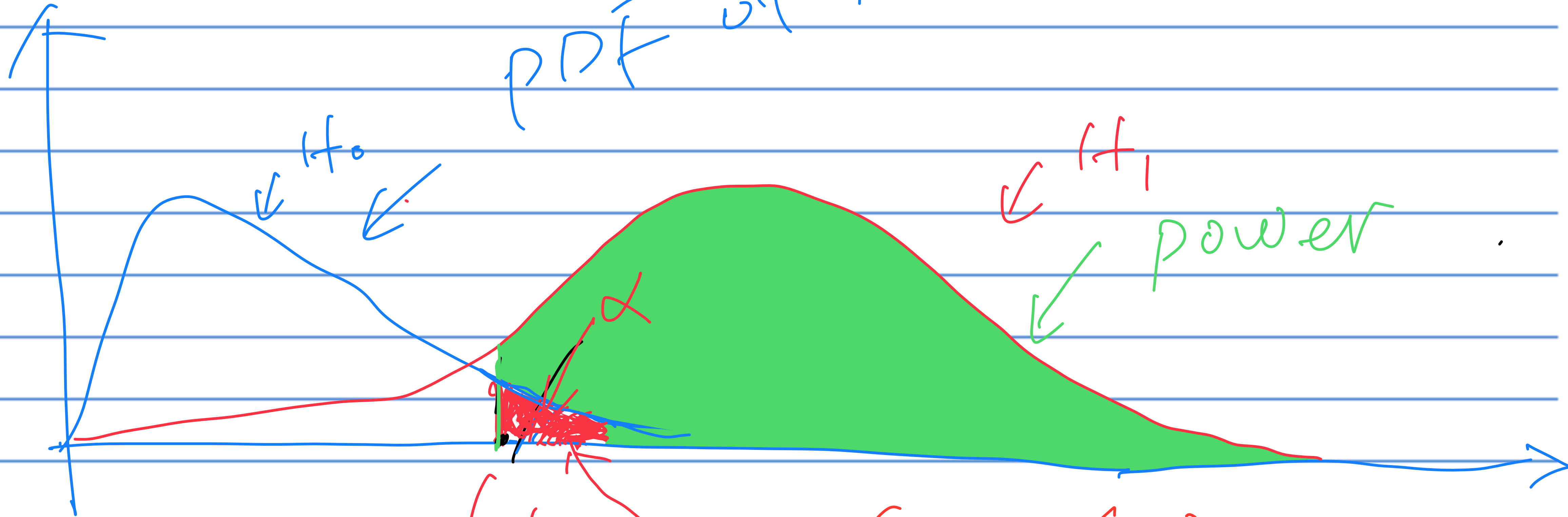
Example of hypothesis testing

y_i	x_i	z_i
y_1	x_1	0
y_2	x_2	0
$\vdots$	$\vdots$	$\vdots$
y_i	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$
y_n	x_n	1

$$H_0 : \begin{matrix} \text{Corr}(Y, X) = 0 \\ \text{Corr}(Y, Z) = 0 \end{matrix} \quad \Bigg| \quad \begin{matrix} H_1 : \\ \text{Corr}(Y, X) \neq 0 \end{matrix}$$

Hypothesis test

PDF of T under H_0



critical value

$$P(T > c_\alpha) = \alpha$$

Two types of errors.

	$T > C_\alpha$	$T \leq C_\alpha$
H_0	Type 1 error (FP)	TN
H_1	TP	Type 2 error (FN)

$$P(\text{FP}) = \hat{P}(T > C_\alpha | H_0) = \alpha$$

$$\text{power} = P(\text{TP}) = \hat{P}(T > C_\alpha | H_1)$$

power is also called Sensitivity

Under

ifo

$T^{(i)}$

C_α

$T^{(i)} > C_\alpha$

$T^{(1)}$

C_α

$P(T > C_\alpha | H_0)$

$T^{(2)}$

C_α

1

$\sum_{i=1}^N I(T^{(i)} > C_\alpha)$

$\vdots$

$\vdots$

$=$

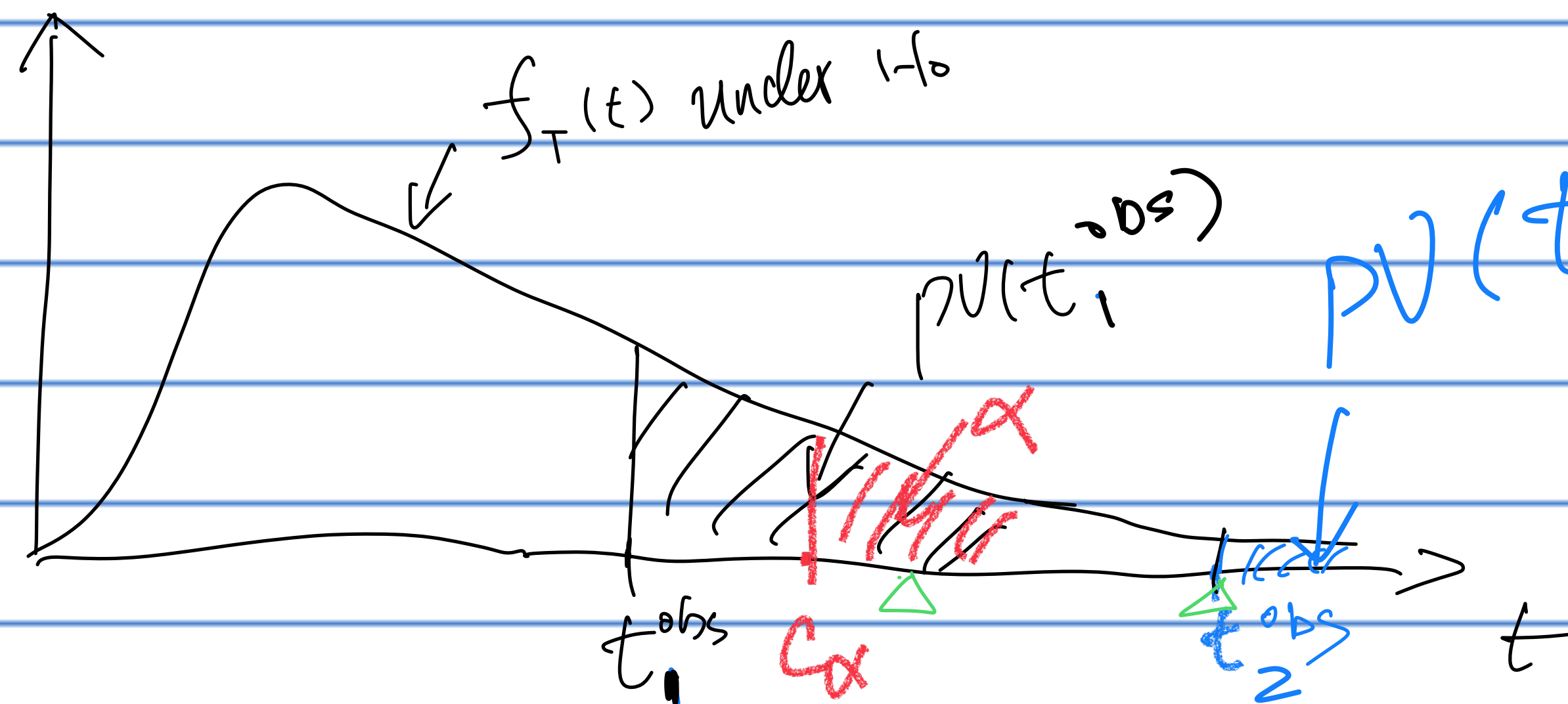
$\frac{\alpha}{N}$

$T^{(N)}$

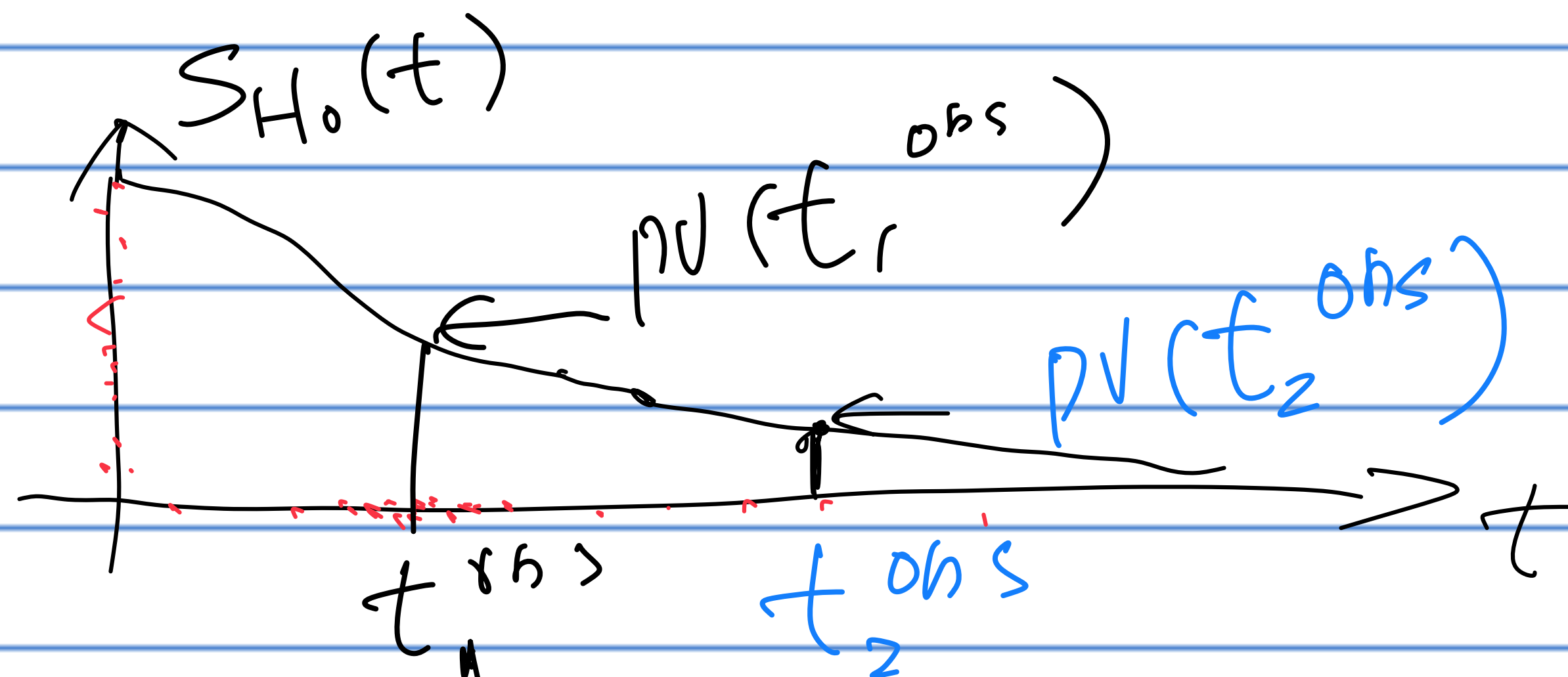
C_α

0

p-value (Definition)



$$\begin{aligned}
 P(t) &= P(T > t | H_0) \\
 &= 1 - F_{H_0}(t) \\
 &= S_{H_0}(t)
 \end{aligned}$$



Uniformity of $PV(T)$ under H_0

$$PV(T) | H_0 \sim \text{Unif}(0, 1)$$

Pf:

$PV(T) = 1 - F_{H_0}(T)$, where F_{H_0} is the CDF of $T | H_0$
Under H_0 :

$$F_{PV(T)}(t) = P(PV(T) < t) = P(1 - F_{H_0}(T) < t)$$

$$= P(F_{H_0}(T) > 1 - t)$$

$$= P(T > F_{H_0}^{-1}(1 - t))$$

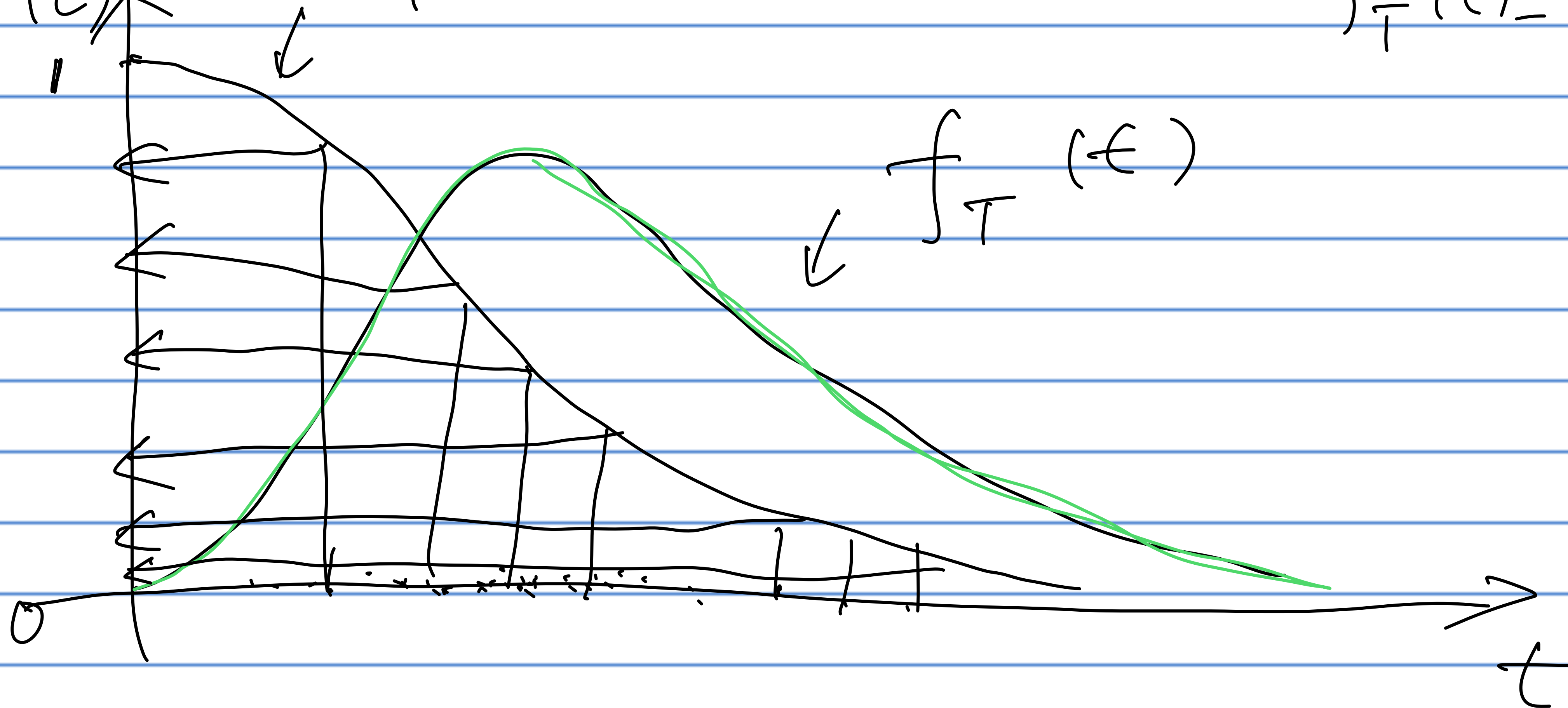
$$= 1 - F_{H_0}(F_{H_0}^{-1}(1 - t)) = 1 - (1 - t)$$

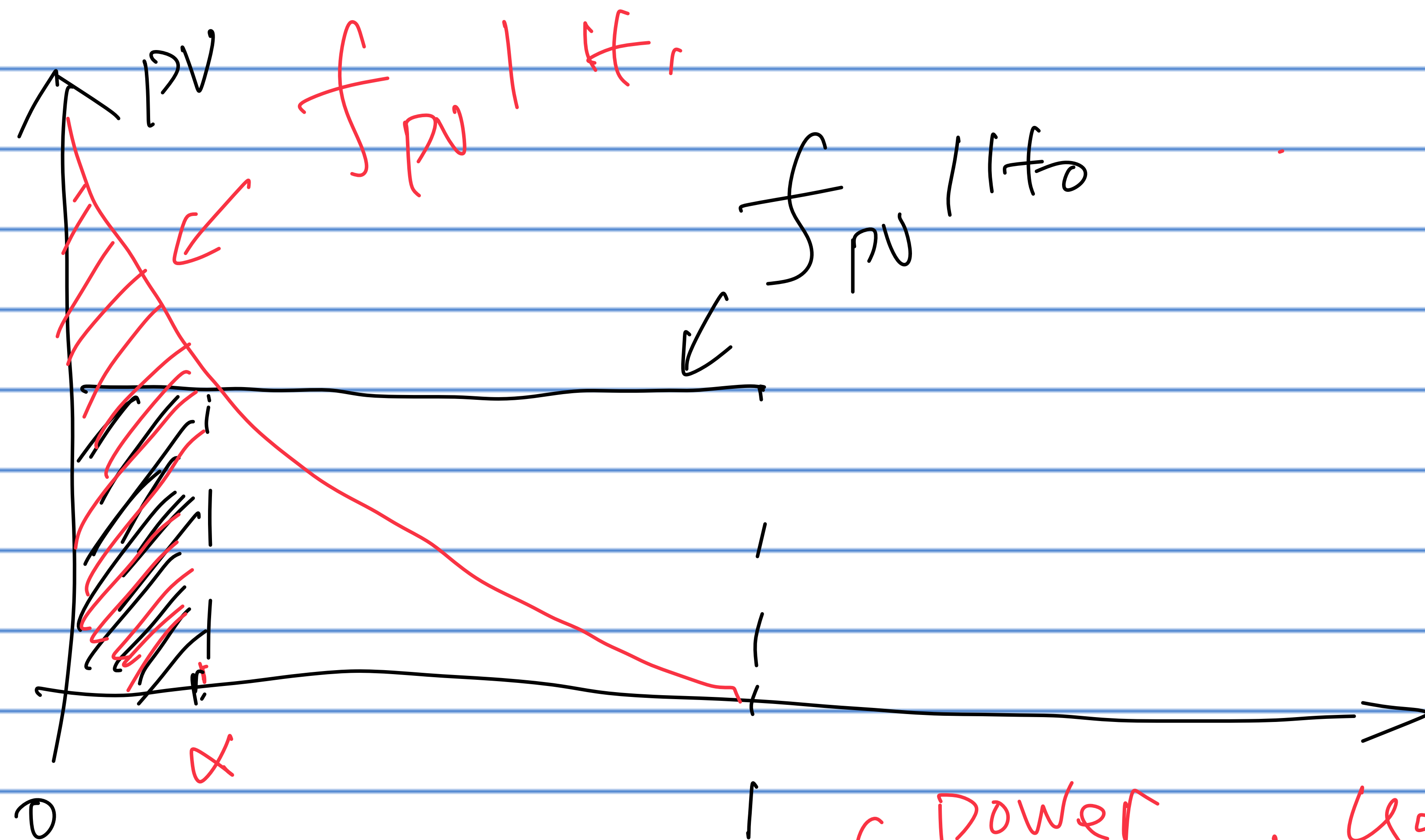
$$= t$$

under H_0
 $p_N(t)$

$$S_T(t) = 1 - F_T(t)$$

$$f_T(t) = -S_T'(t)$$





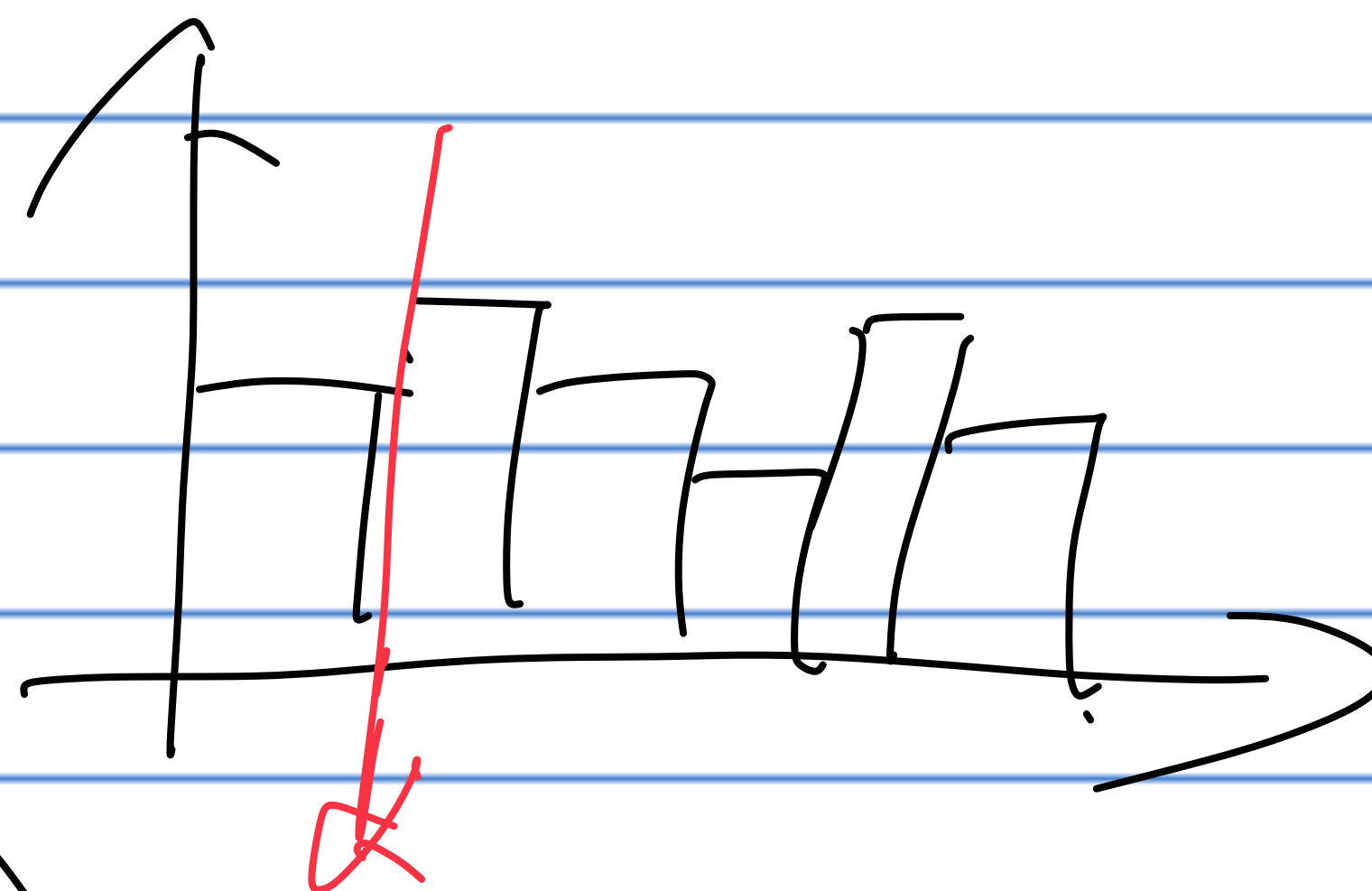
$$P(PV < \alpha) = \begin{cases} \text{power, under } H_1 \\ P(\text{False Positive}), \text{ under } H_0 \end{cases}$$

Under (f)

$$D^{(1)} \rightarrow T^{(1)} \rightarrow \hat{P}_V^{(1)}$$

⋮

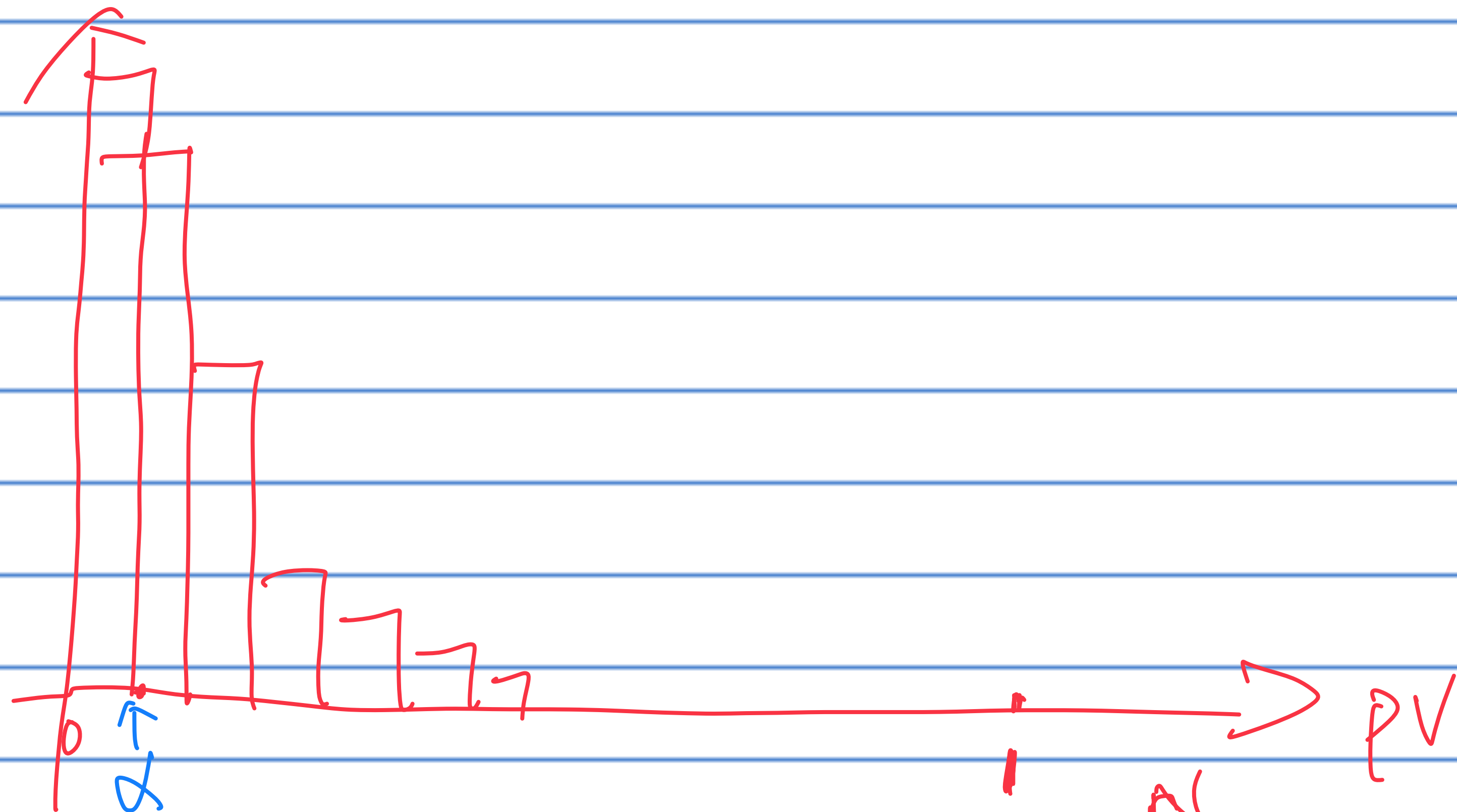
$$D^{(N)} \rightarrow T^{(N)} \rightarrow \hat{P}_V^{(N)}$$



$$\hat{P}\left(\frac{1}{N} \sum_{i=1}^N \mathbb{I}(\hat{P}_V^{(i)} < \alpha)\right)$$

N

Under H_1



$$\hat{P}(PV < \alpha) = \frac{\sum_{i=1}^N I(PV^{(i)} < \alpha)}{N}$$

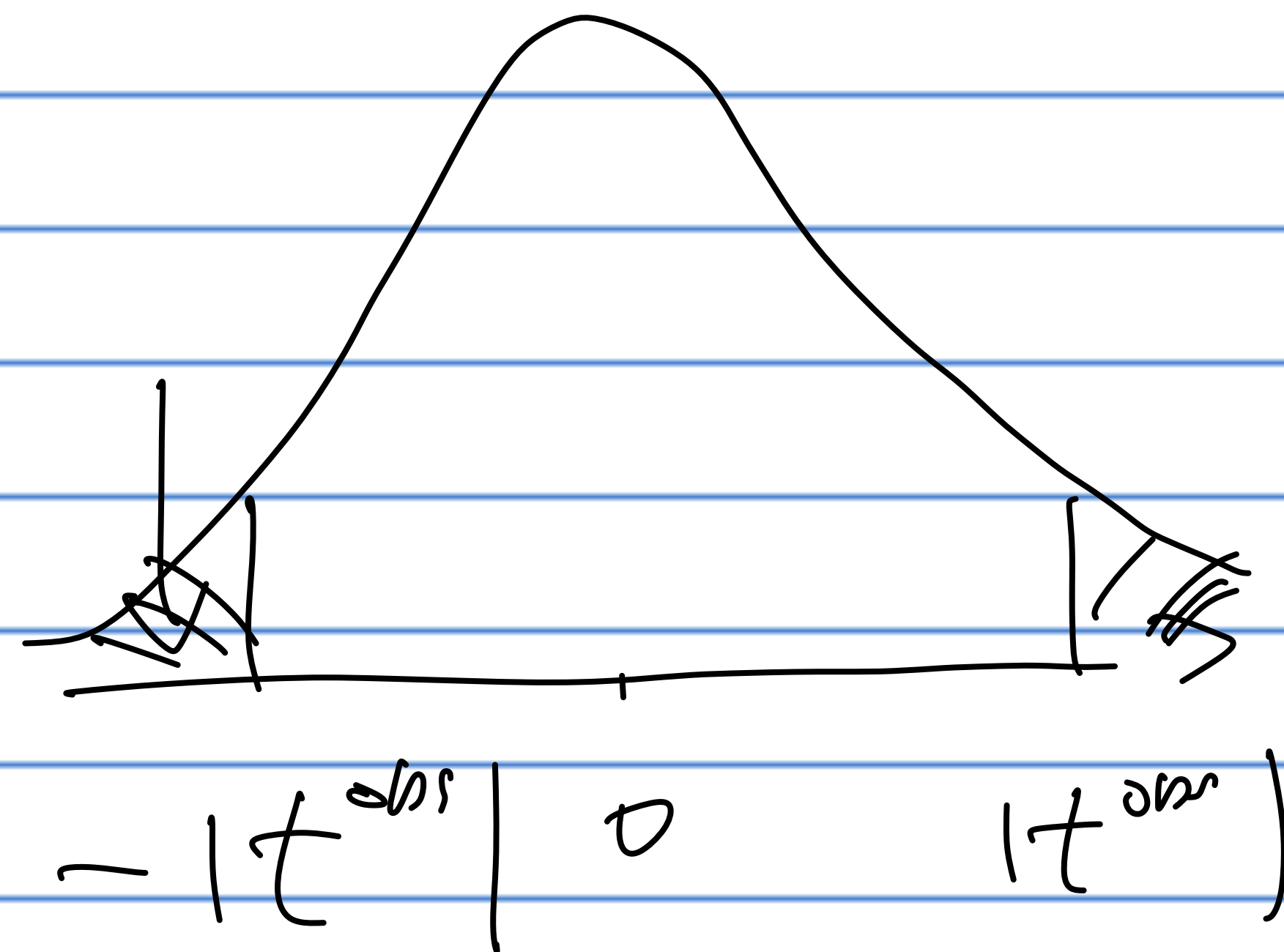
Example:

$$x_1, \dots, x_n \sim N(\mu, \sigma^2)$$

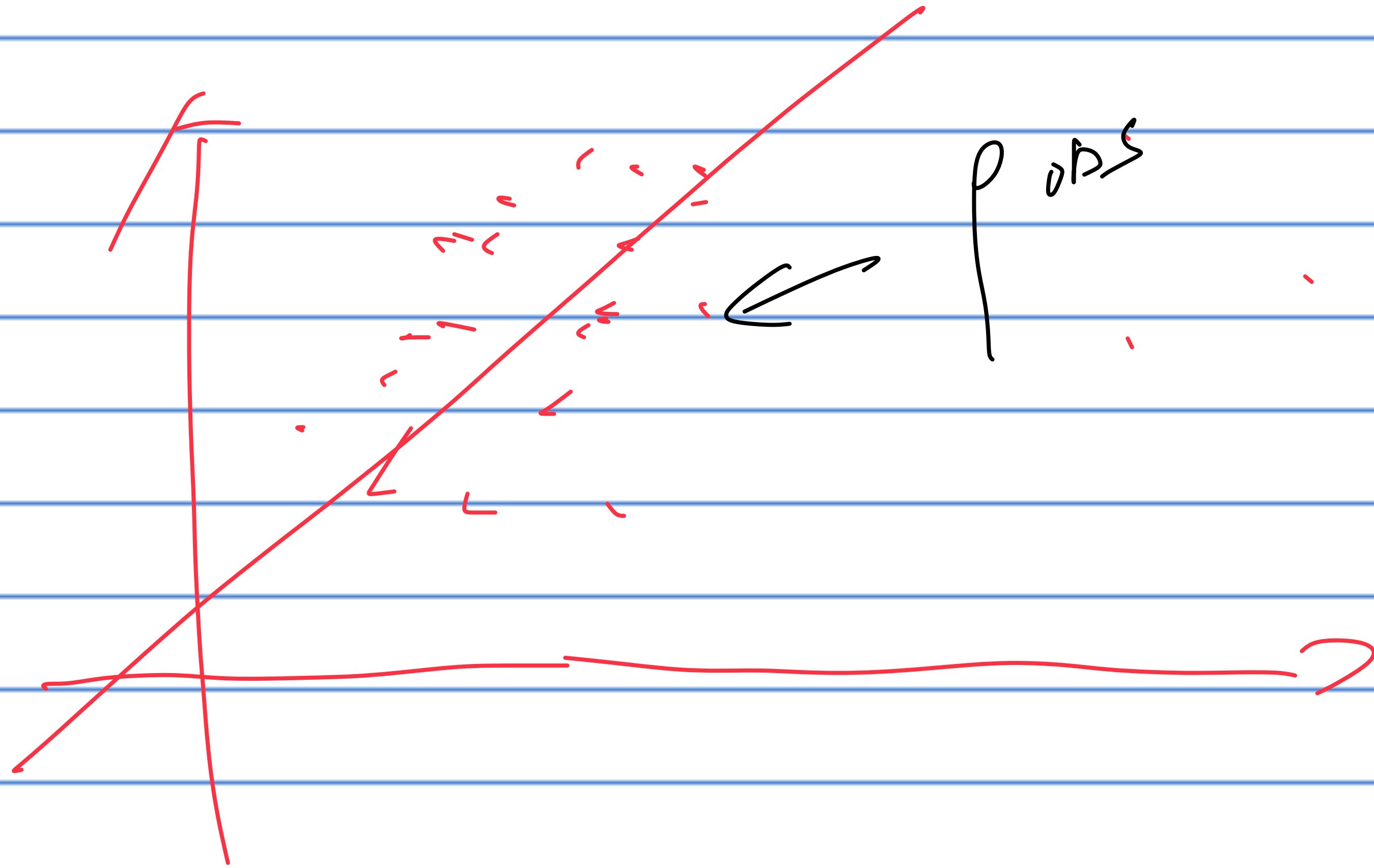
$$T = \frac{\bar{x} - 0}{\sigma/\sqrt{n}}$$

$$H_0: \mu = 0 \text{ vs } H_1: \mu \neq \mu_0$$

$$pV(T | H_0) = \sim \overset{t_{n-1}}{P}(T > |t^{obs}|) \times 2$$



Monte Carlo for calculating $pV(t)$



x	y
10	9
1	2
2	3
1	2

✓

obs

$$pV(t^{obs}) = P(\rho > \rho^{obs} | H_0)$$

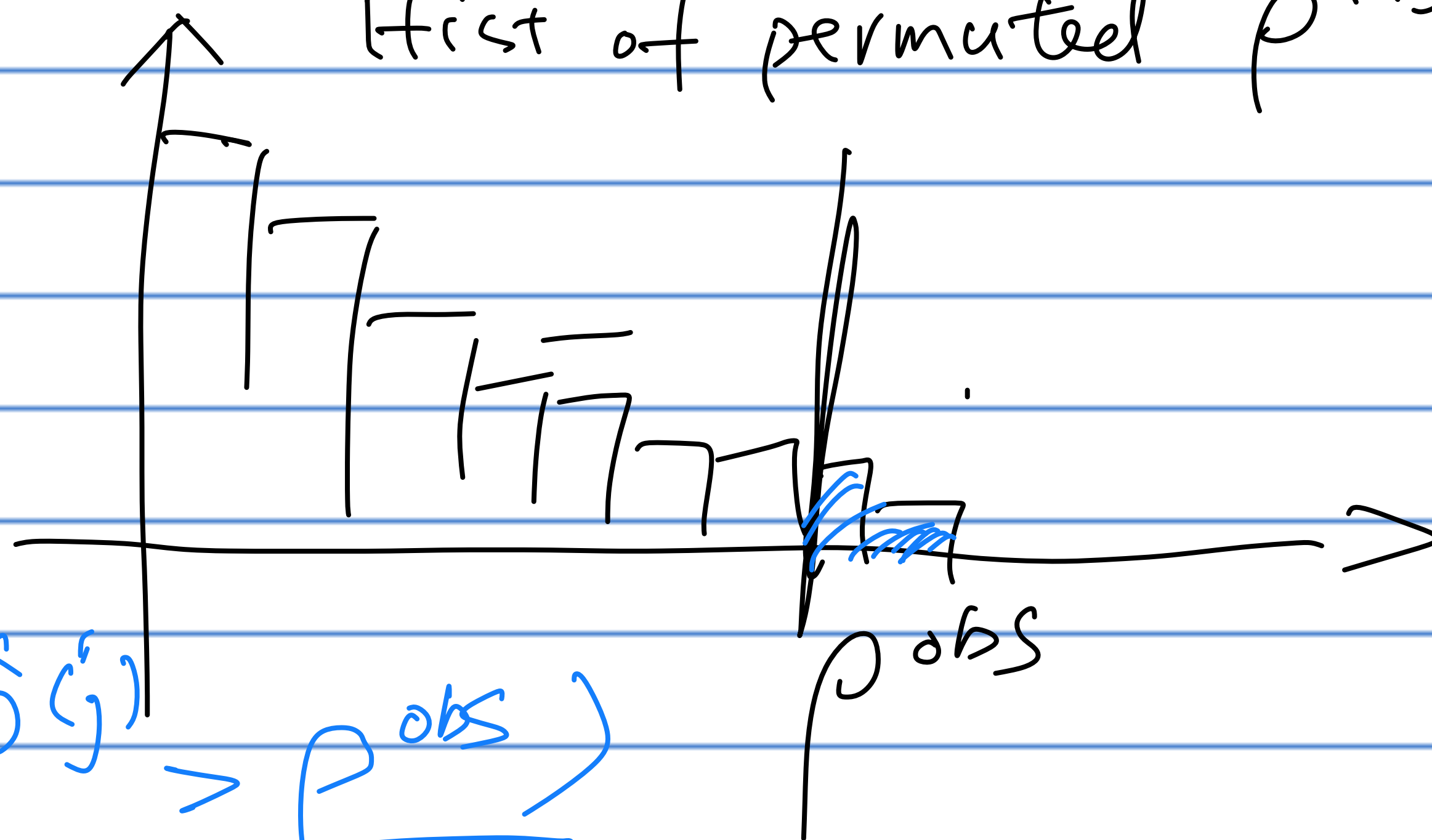
Permutation test
 $H_0: x_i \text{ \& \& } y_i \text{ are independent}$

x_i	y_i
x_1	y_1
$\vdots$	$\vdots$
x_n	y_n

Under H_0 , $\rho \approx 0$

Under H_1 , $\rho \neq 0$

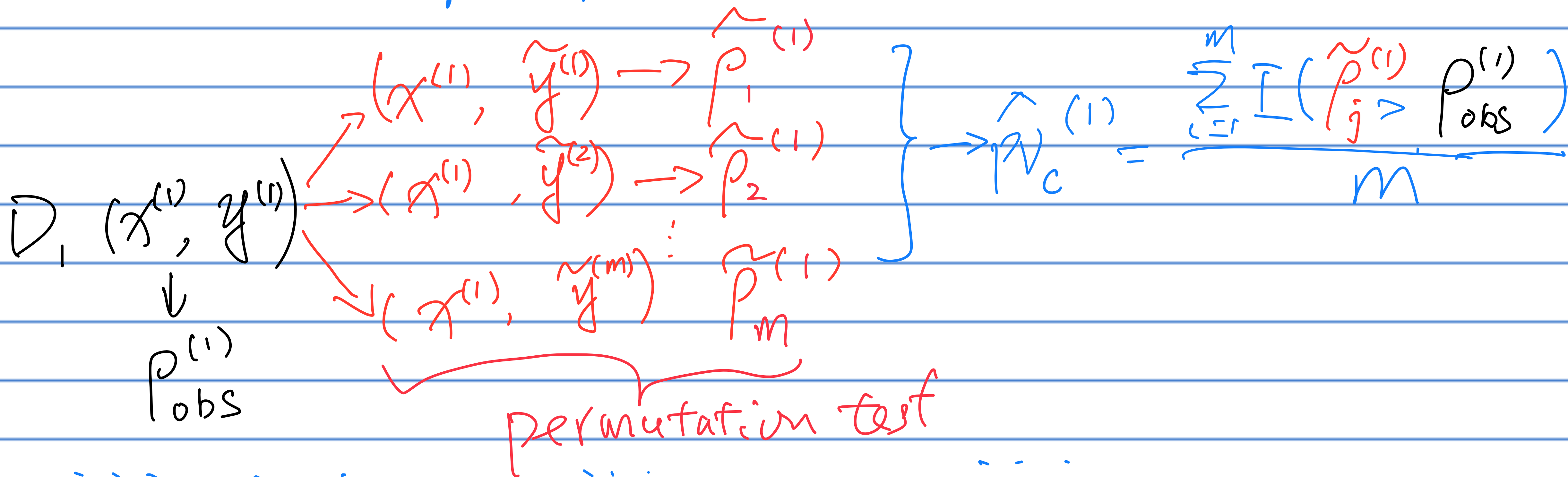
Hist of permuted $\hat{\rho}^{(i)}$



$$\hat{p}_N(\rho^{obs}) = \frac{\sum_{j=1}^m \mathbb{I}(\hat{\rho}^{(j)} > \rho^{obs})}{N}$$

Evaluate

$\hat{p}_V(p_{obs})$:



...

Observation $D_N = (x^{(N)}, y^{(N)})$ is compared to the observed p-value $p^{(N)}_{obs}$.

The permutation test involves generating m permuted data points $(x^{(N)}, y^{(N)}) \rightarrow (x^{(N)}, y^{(1)}) \rightarrow p^{(N)}$, $(x^{(N)}, y^{(N)}) \rightarrow (x^{(N)}, y^{(2)}) \rightarrow p^{(N)}$, ..., $(x^{(N)}, y^{(m)}) \rightarrow p^{(N)}$.

The permutation test statistic is calculated as:

$$\hat{p}_V^{(N)} = \frac{\sum_{i=1}^m I(\tilde{p}^{(N)}_i > p^{(N)}_{obs})}{m}$$

power linear models

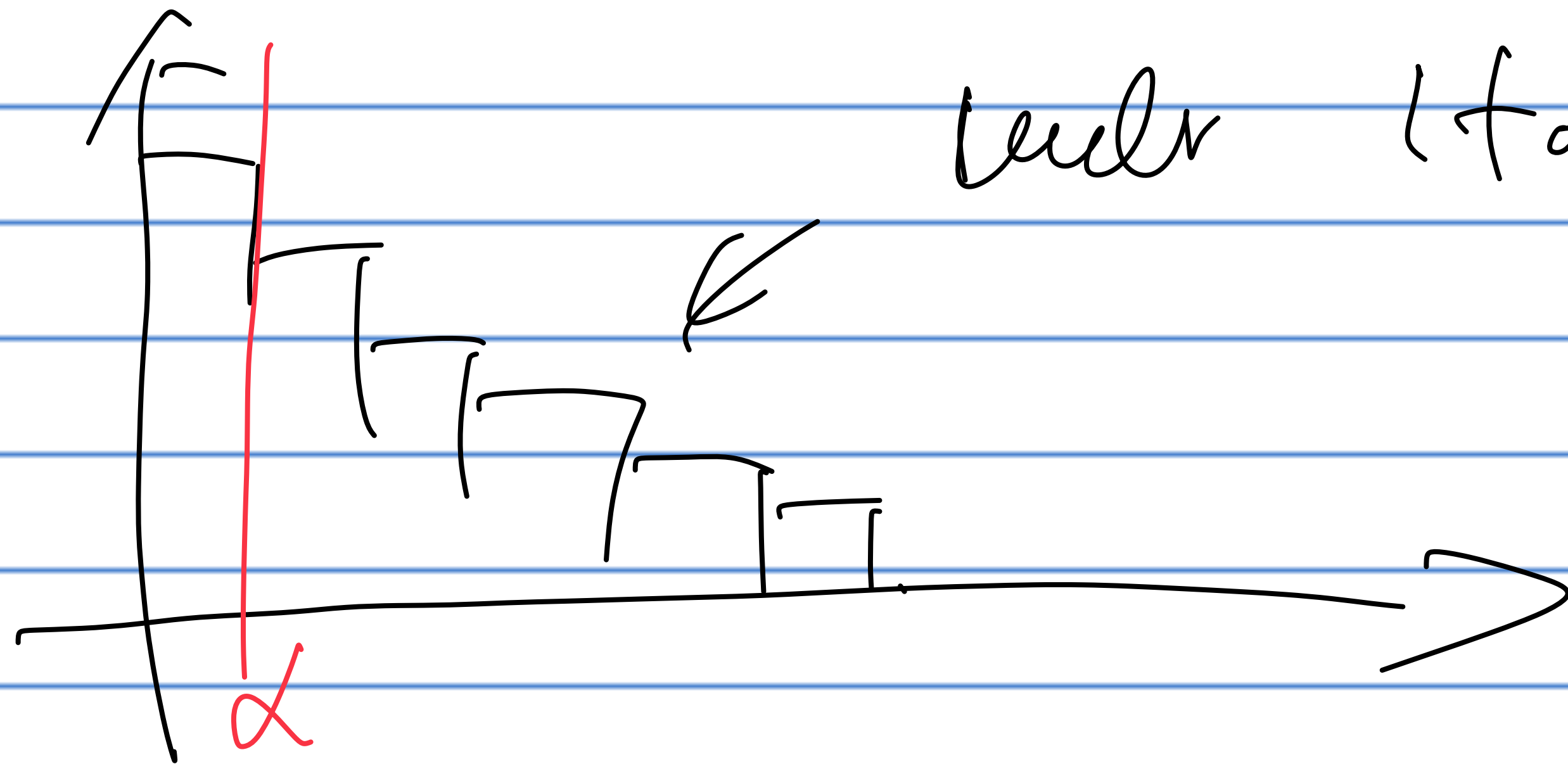
y_i	x_i	$x_i^{\frac{1}{2}}$	$x_i^{\frac{3}{2}}$	...	x_i^2
x	x				
x	x				
.	.				
.	.				
.	.				
x	x				

$|r^{(\frac{1}{2})}|$
 $|r^{(\frac{3}{2})}|$
 $|r^{(2)}|$

power selection

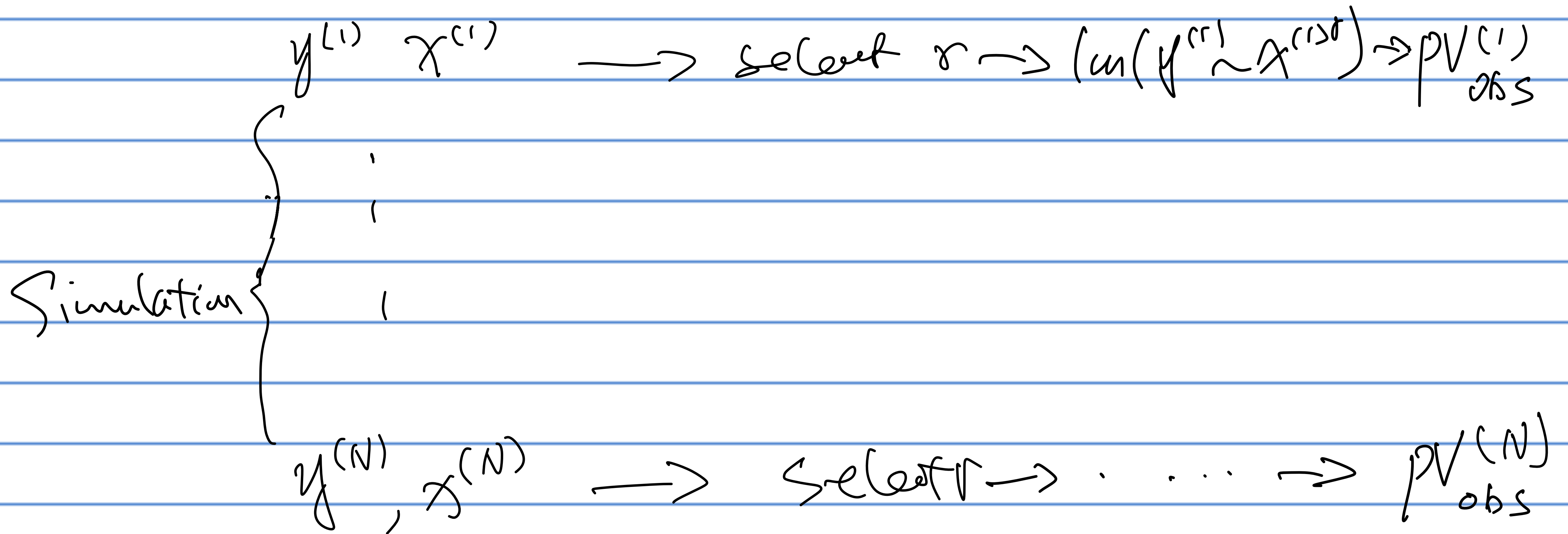
Naive test:

$y, x \rightarrow \text{select } x^r \text{ with } p \rightarrow \ln(y \sim x^r) \rightarrow p\text{-value}$



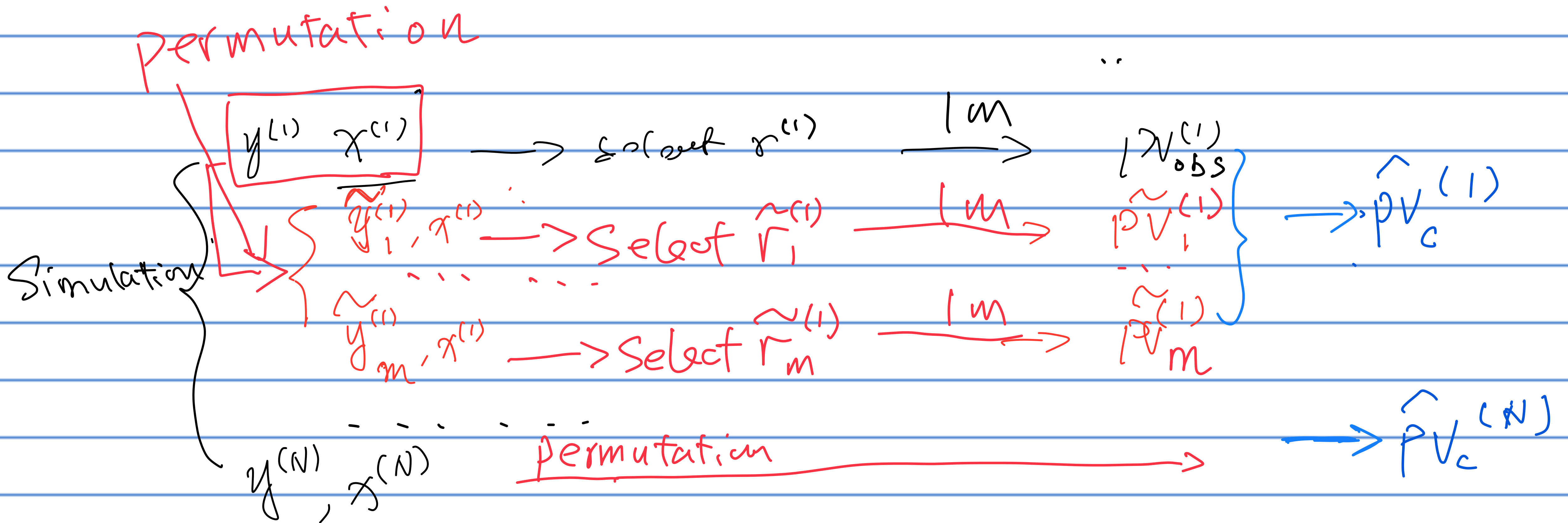
$\hat{p}(p\text{-value} < \alpha) > \alpha$ (biased)

Simulation for evaluating naive test (power selection)



Note: This naive procedure is incorrect, producing higher probability of false positive.

Permutation test

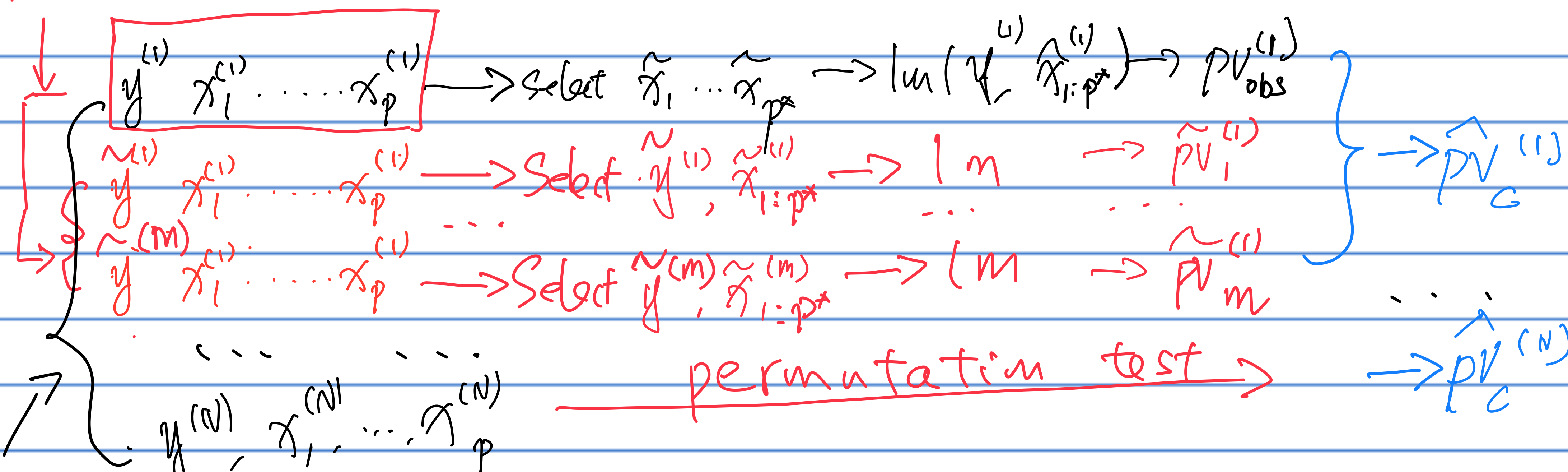


$$\hat{pV}_c^{(i)} = \frac{1}{m} \sum_{j=1}^m \mathbb{I}(pV_j^{(i)} < pV_{\text{obs}}^{(i)})$$

where $m = \#$ of m random permutations

Question 1 of Assignment 1

permutation



Simulation

$$\hat{pV}_c^{(i)} = \frac{\sum_{j=1}^m I(\hat{pV}_j^{(i)} < pV_{obs}^{(i)})}{m}, \text{ for } i = 1, 2, \dots, N$$

$pV_{obs}^{(i)}$: naive test (incorrect p-values)

$\hat{pV}_c^{(i)}$: corrected test (correct p-values)